

----- Active space -----

**Salivary Gland Tumors**

Larger glands : m/c benign tumors.

Smaller glands : m/c malignant tumors.

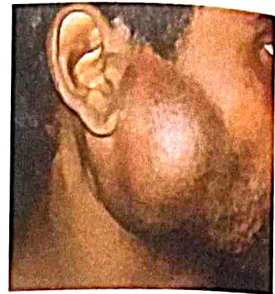
Tumors of salivary glands :

|               | % Benign | % malignant |
|---------------|----------|-------------|
| Parotid       | 90       | 10          |
| Submandibular | 50       | 50          |
| Sublingual    | 20       | 80          |
| minor         | 10       | 90          |

**Parotid tumors :**

Features :

- Swelling on the side of the face.
- Lift the ear lobule.
- m/c : Superficial lobe.
- Pain d/t stretching of capsule/fascia.
- Deep lobe enlargement : Tonsillar fossa pushed medially.



Parotid tumors

Investigation :

- CT/MRI.
- IOC : FNAC.

m/c parotid tumor —→ Benign : Pleomorphic adenoma.  
 —→ Children : Hemangioma.  
 —→ m/c malignant : mucoepidermoid carcinoma > Adenoid cystic carcinoma.

**BENIGN TUMORS****Pleomorphic adenoma :**

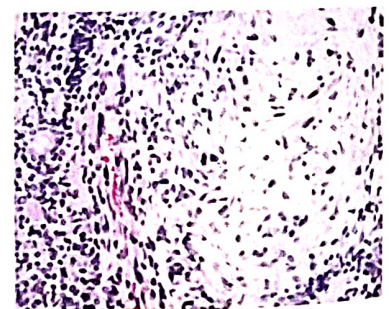
- m/c tumor of parotid gland.
- m/c in female, 5<sup>th</sup> decade (3<sup>rd</sup> - 6<sup>th</sup> decade).

Clinical features :

- Slow growing parotid swelling.
- Commonly involve superficial lobe.

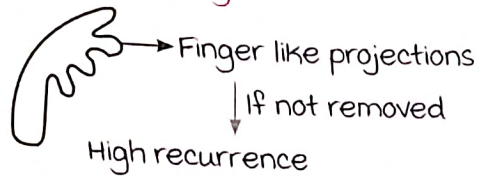
Investigations :

- CT/MRI.
- IOC : FNAC.
- HPE : Epithelial/myoepithelial component in a myxoid background (Triphasic tumor).
- IHC : CK7 (Strong & diffuse pattern).
- myoepithelial cells express p63, S-100, SOX10, SMA.



HPE of pleomorphic adenoma

management : Superficial parotidectomy.



----- Active space -----

mixed malignant tumor/Carcinoma ex-pleomorphic adenoma :

- Long standing case → malignant transformation.
- Features s/o malignancy.
  - Rapid ↑ in size.
  - Painful.
  - Facial nerve involvement.
  - Ulceration.
  - Lymph node involvement.
- IOC : FNAC.
- management : Sx Followed by Radiotherapy.

Warthin's tumor / Adenoma lymphomatosum :

- 2<sup>nd</sup> m/c parotid tumor.
- Strong association with smoking, radiation exposure.

Clinical features :

- m/c in male, 6<sup>th</sup> decade.
- Exclusively seen in parotid.
- 10% : B/L.
- Present as a painful parotid swelling.
- malignant transformation : Rare.

Investigations :

- Imaging : CT/MRI.
- IOC : FNAC.

Two layers of eosinophilic cells rich in mitochondria

Lymphocytic infiltration (+)



HPE of Warthin's tumor

management : Superficial parotidectomy.

## MALIGNANT TUMORS

Mucoepidermoid carcinoma :

- m/c salivary gland malignancy in children and young adults.
- Peak incidence : 2<sup>nd</sup> decade.

Risk factor : Radiation/Chemotherapy during childhood.

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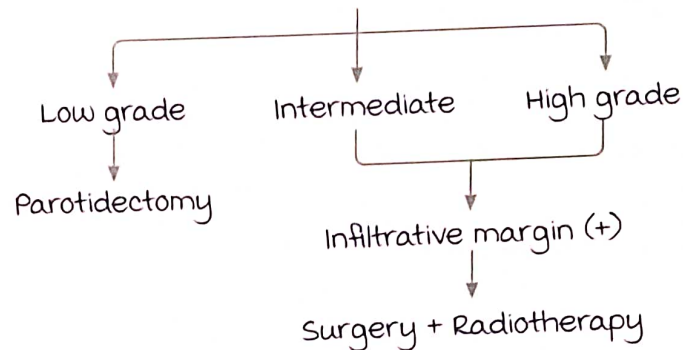
## Clinical features :

- Fast growing parotid swelling (+).
- Facial nerve maybe involved.

## Investigations :

- Imaging : CT/MRI.
- IOC : FNAC.

## management :



## Adenoid cystic carcinoma :

2<sup>nd</sup> m/c malignant parotid tumor.

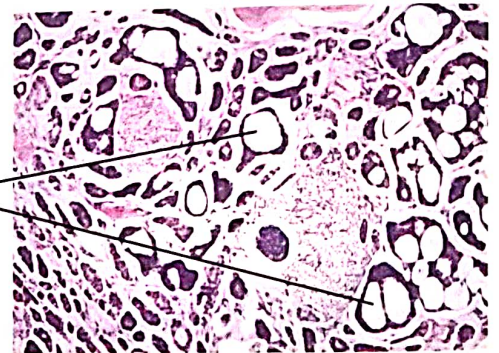
## Clinical features :

- Present as parotid swelling.
- Pain (+) d/t perineural invasion.

## Investigation :

- IOC : FNAC.
  - HPE : Swiss cheese pattern.
  - IHC — { Ductal cells : ckit (+)  
myoepithelial cells : p63 and SMA.

Swiss cheese pattern



HPE of adenoid cystic carcinoma

management : Surgery + Radiotherapy.

## Treatment principles for malignant tumors :

- margin : 0.5 cm.
  - T3/T4 tumors.
  - High grade tumors.
- } Elective SOHND

## Adjuvant RT :

1. Stage 3 & 4 tumors.
2. High grade.
3. Positive margins.
4. PNI/LVI.
5. Extra nodal spread.

## Index :

SOHND : Supra Omohyoid Neck Dissection

LVI : Lymphovascular invasion

PNI : Perineural invasion

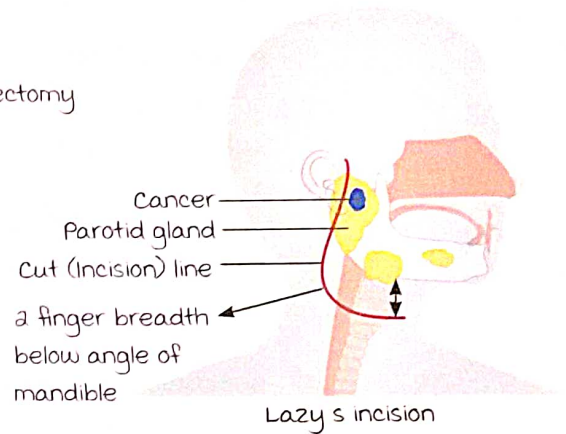
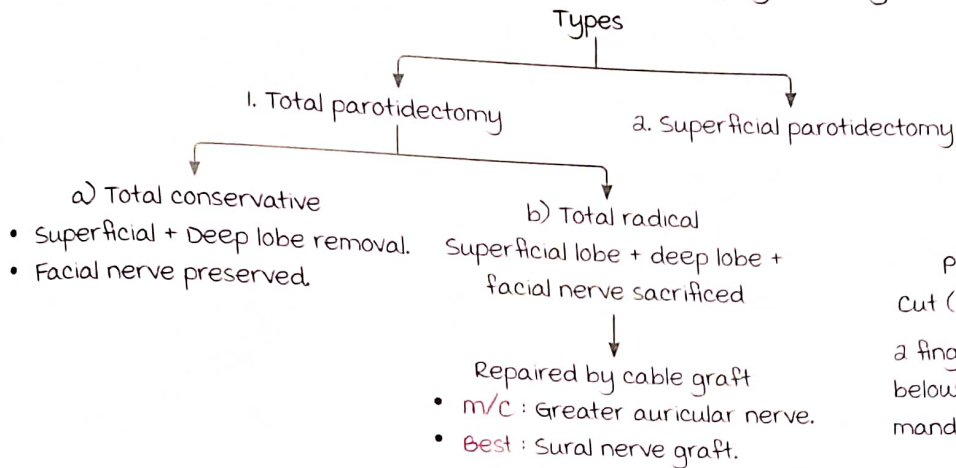
RT : Radiotherapy

## Surgery For Parotid Gland

00:44:55

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Incision : **Lazy s**/modified Blair's incision (Prevent injury to marginal mandibular nerve).

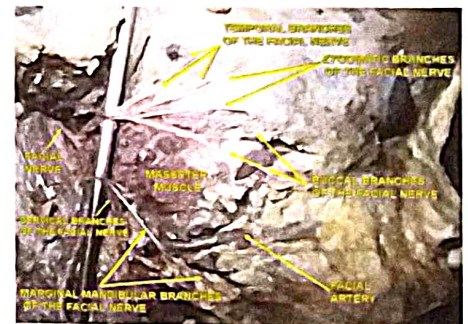


## COMPLICATIONS

1. Hemorrhage.

2. Nerve Injury.

- marginal mandibular nerve/Ramus marginalis → **Drooping of mouth.**
- Greater auricular nerve → **Anaesthesia over beard region.**
- Facial nerve injury : Prevented by pointers.
  - a. **Tragal** (most specific) : Nerve lies 1 cm inferior & deep.
  - b. Styloid process.
  - c. Posterior belly of digastric.
  - d. Retrograde method.
  - e. Nerve stimulators.



Branches of facial nerve

3. Fistula
- Low output (with gland) → Closes spontaneously.
  - High output (with duct) → **Newman & Seabrook surgery.**

4. Recurrence.

5. **Frey's syndrome** : Gustatory sweating.

- Nerve misdirected : Auriculotemporal nerve (post ganglionic parasympathetic fibres).
- Starch iodine test : True colour confirms diagnosis.
- management :
  - Botox
  - Anti-perspirants.
  - Tympanic neurectomy (Definitive).
- Prevention : muscle flap to cover parotid (Sternocleidomastoid/digastric).



Starch iodine test

----- Active space ----- Note :

Extracapsular dissection :

- Done for benign tumors.
- No effect on oncological safety (Recurrence).
- Less incidence of facial nerve injury & Frey's syndrome.

## Submandibular and Sublingual Glands

00:01:15

Submandibular swelling :

- Bimanual palpation : To differentiate from lymph nodes.

Submandibular swelling : Palpable

Lymph nodes : Not palpable

- m/c tumor — Benign : Pleomorphic adenoma.
- malignant : Adenoid cystic carcinoma.

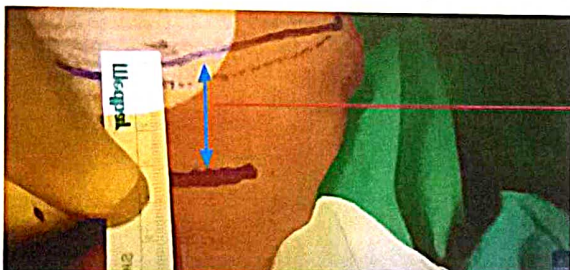
Investigation :

IOC : FNAC.

management : Submandibular gland excision.

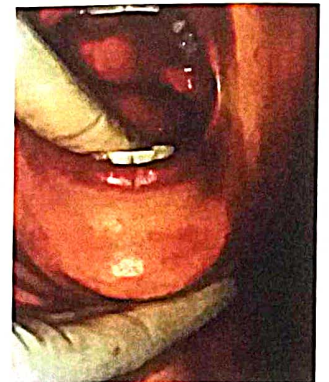


Submandibular swelling



a finger breadth below angle of mandible

Prevent injury to marginal mandibular nerve



Bimanual palpation

Complications of submandibular gland surgery :

1. Hemorrhage.

2. Nerve injury — Lingual nerve

                              — marginal mandibular nerve : m/c

                              — Hypoglossal nerve

3. Injury to other structures

- Anterior facial vein.
- Facial artery.

Sublingual & minor salivary gland :

- m/c tumor : Adenoid cystic carcinoma.
- management : Excision + Radiotherapy.

# SKIN TUMORS AND SOFT TISSUE SARCOMAS

----- Active space -----

## Basal Cell Carcinoma (BCC)

00:01:50

AKA **Rodent Ulcer**.

Characteristic features :

- Locally invasive.
- mets (-) → **Lymph node involvement (-)**.  
→ Distant mets (-).
- m/c site : **Face** (Above line joining angle of mouth to ear lobule).

Risk factors :

- White population.
- UV rays.
- Gorlin syndrome : Chromosome 9.

Types :

- |                          |                 |                  |
|--------------------------|-----------------|------------------|
| • Localized :            | • Generalized : | • Infiltrative : |
| - Nodular : <b>m/c</b> . | - Superficial.  | morhoeic : most  |
| - Nodulocystic.          | - Multifocal.   | aggressive.      |
| - Pigmented.             |                 |                  |

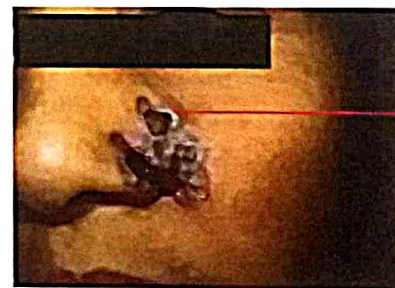
Clinical features :

Ulcer : **Rolled out, pearly white edges**.

Investigations :

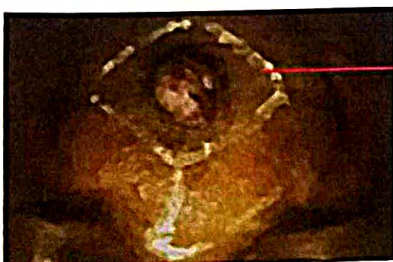
Biopsy : **Pallisading pattern**.

management : **Wide local excision + Rhomboid/Limberg flap**.

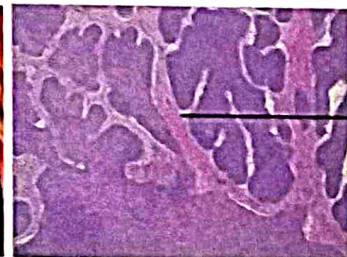


→ Rolled out  
pearly edges

Basal cell carcinoma



→ Rhomboid  
flap



→ Pallisading  
pattern

Biopsy of BCC

- BCC close to tip of nose : **Bilobed flap**.
- BCC involving eyelid region : **Bipedicled flap**.

----- Active space -----



Bilobed flap

**High risk features :**

- > 2 cm size.
- Sites where infiltration can lead to cranial extension.
- Recurrent lesions.
- After immunosuppression.

**Moh's micrographic surgery :**

- Layer by layer removal of lesion + microscopic examination.
- m/c done for BCC (Also used for other skin cancers).

**Indications :**

- Recurrent cases.
- Close to vital structures.
- Close to vital nerves.

**Advantages :**

- Less tissue removed.
- Better cosmesis.

Disadvantage : Time consuming.

**Malignant Melanoma & Squamous Cell Carcinoma**

00:03:16

**MALIGNANT MELANOMA**

Tumor of the melanocytes.

**Risk factors :**

- UV radiation.
- White population.
- Familial atypical mole melanoma syndrome (Also ↑ risk of pancreatic cancer).

**Phase of growth :**Horizontal Followed by vertical invasion.

malignant melanoma.

**Types :****I. Superficial spreading :**

- m/c type.
- Prolonged horizontal phase.
- m/c melanoma in pre-existing mole.

**Risk factors :**

- Young individuals.
- Sun exposure.

## 2. Lentigo maligna :

- AKA Hutchinson's melanotic freckle.
- In-situ melanoma.
- Elderly patients.
- Best prognosis.

## 3. Acral melanoma :

- m/c melanoma in **dark skinned individuals**.
- Site : Palm & sole.
- Rapid vertical phase.
- Poor prognosis.

## 4. Nodular melanoma :

- Rapid vertical phase.
- Worst prognosis.

## Variant : Amelanotic melanoma.

- Lacks pigment.
- Detected late.

## 5. Desmoplastic melanoma :

- Site : Head & neck region.
- Perineural invasion (PNI) (+) → Painful.  
→ Locoregional recurrence (+).

## Features suggesting malignant transformation :

## ABCDE changes :

- **A**symmetry.
- **I**rrregular **b**order.
- **C**olour change.
- **D**iameter >6 mm.
- **E**volution.

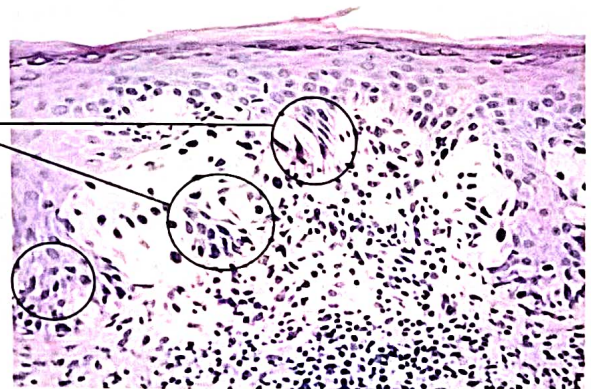
## Investigations :

## 1. Biopsy.

## 2. IHC markers :

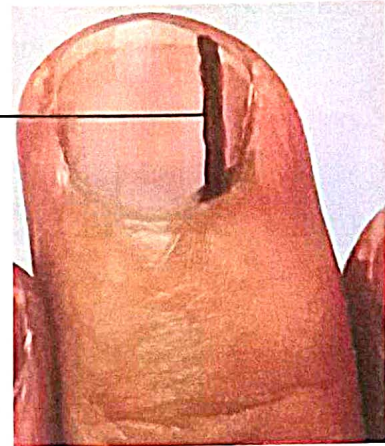
- a. S-100.
  - b. melan-A.
  - c. HMB 45.
- } **Brown color (+) : Positive.**

melanocytes in  
epidermis & dermis



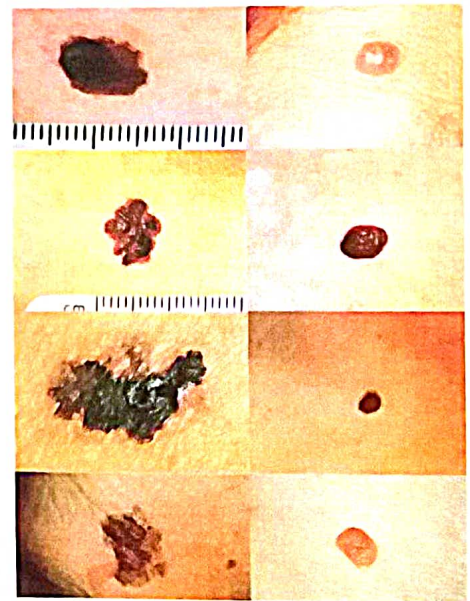
Biopsy of malignant melanoma

Hutchinson's  
sign



Subungual melanoma

m/c type : Superficial spreading.



ABCDE changes of melanoma

----- Active space -----

**Staging :**

Based on depth of invasion.

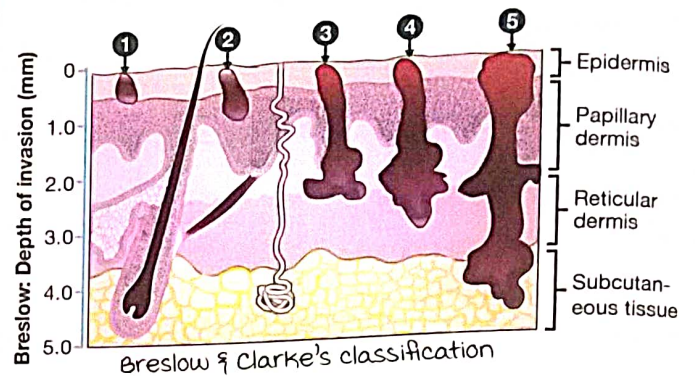
1. Clarke's classification : Based on **structure** invaded.

| Stage | Structure involved   |
|-------|----------------------|
| I     | Epidermis.           |
| II    | Papillary dermis.    |
| III   | Dermal papillae.     |
| IV    | Reticular dermis.    |
| V     | Subcutaneous tissue. |

2. Breslow classification :

Based on **size**.

| Stage | Size           |
|-------|----------------|
| I     | $\leq 0.75$ mm |
| II    | 0.76 - 1.5 mm  |
| III   | 1.5 mm - 4 mm  |
| IV    | $> 4$ mm       |



Spread :

**management :**

Localised Disease

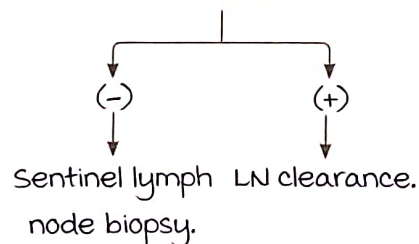
Surgery

1° tumor :

- Wide local excision (WLE).
- **NCCN criteria** for margin.

| Size       | margin   |
|------------|----------|
| $< 0.5$ mm | 0.5 cm   |
| 0.5-1 mm   | 0.5-1 cm |
| 1 mm-2 mm  | 1-2 cm   |
| $> 2$ mm   | 2 cm     |

Gross lymph node (LN) involvement



metastasis

- Dabrafenib : BRAF Inhibitor.
- Trametinib : Against MAPK pathway.
- Ipilimumab : Check point inhibitor.

- Isolated limb perfusion chemotherapy : Locally advanced malignant melanoma ----- Active space -----
- most important prognostic factor
  - Overall : Lymph node status.
  - LN (-) / Early malignant melanoma : Depth.

## SQUAMOUS CELL CARCINOMA

precursor lesions:

- Cutaneous horn.
- Keratoacanthoma.

- Solidified keratin
- Complication of sebaceous cyst



Sebaceous horn

marjolin's ulcer :

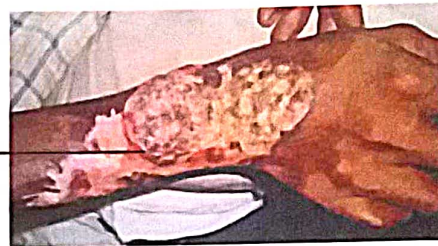
seen in :

- Long standing varicose ulcers.
- Burn scar.

Lymph node: Rarely involved d/t scarring of lymphatics.

Investigation : Biopsy.

management : wide local excision.



marjolin ulcer

Raised, everted  
cauliflower edges

## Soft Tissue Sarcomas (STS)

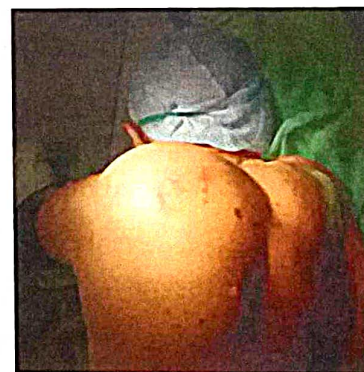
00:24:10

Sarcomas arising from soft tissue, tend to spread via hematogenous route.

m/c STS — Overall  $\uparrow$  in retroperitoneum: **Liposarcoma**.  
In children: **Rhabdomyosarcoma** (Genitourinary  $\uparrow$  head and neck region).

etiology :

- Radiation exposure.
- Chemical exposure.
- Immunosuppression.
- Viral infection : Kaposi sarcoma.
- Long standing lymphedema : Stewart-Treves syndrome (Angio/lymphangiosarcoma).



## Liposarcoma

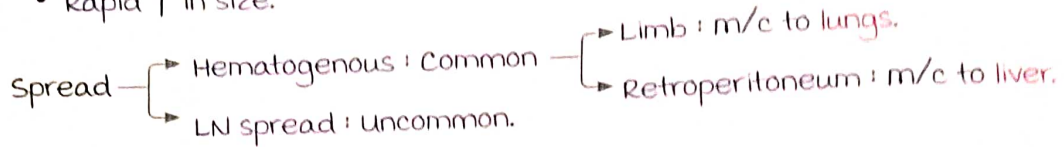
Associated syndromes :

- Retinoblastoma.
- Li-Fraumeni syndrome.
- NF-1.
- Gardner's syndrome : Desmoid tumor.
- Gorlin syndrome : Sarcomas + BCC.

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**Features :**

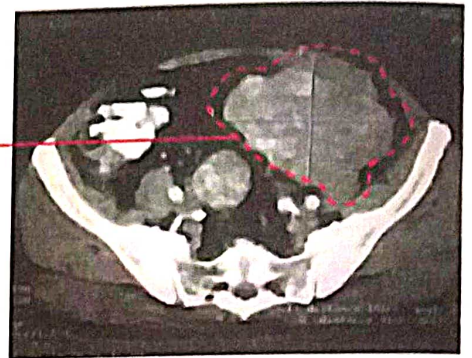
- mass.
- Painless  $\xrightarrow{\text{Nerve infiltration}}$  Painful.
- Swelling.
- Rapid  $\uparrow$  in size.



Sarcomas which spread to lymph nodes :

mnemonic : **MARCES**.

- malignant fibrous histiocytoma.
  - Angiosarcoma.
  - Rhabdomyosarcoma.
  - Clear cell.
  - Epithelial.
  - Synovial sarcoma
- LN clearance required.



Soft tissue sarcoma

**Investigations :**

Core needle biopsy.

Prognostic factors : **Grade** based on mitotic figures (most important).**management :**

1. Surgery : WLE/compartamental excision.
2. Chemotherapy : methotrexate + Adriamycin.
3. Radiotherapy :  $\downarrow\downarrow$  Loco-regional recurrence.

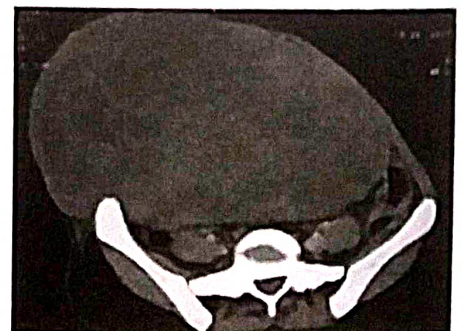
**DESMOID TUMOR**

Site :

- STS over anterior abdominal wall, over scar.
- Association : Gardner's syndrome.

Features :

- Locally invasive.
- Does not metastasize.
- management :
  - WLE.
  - Tamoxifen.



Desmoid tumor

# THORAX AND MEDIASTINUM

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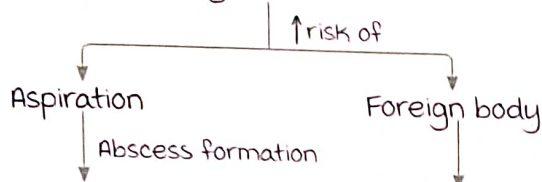
## Surgical anatomy :

Right lung : upper, middle, lower lobe.

Left lung : upper, lower lobe.

Right mainstem bronchus :

- Shorter & straighter :



m/c site :

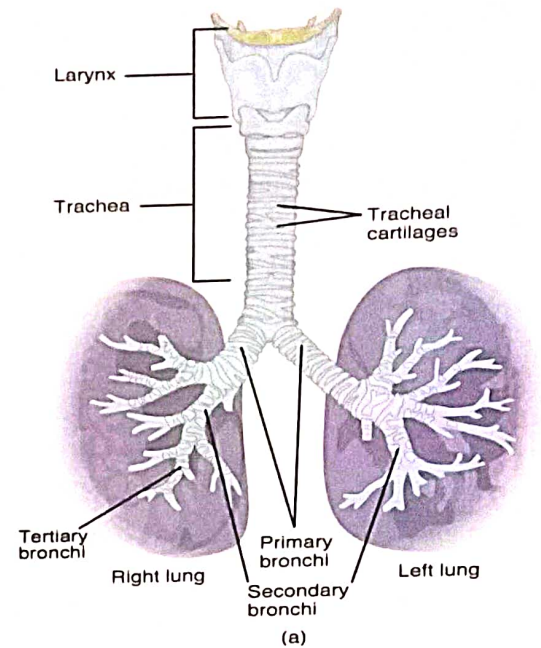
- Posterior upper lobe.
- Superior lower lobe.

Clinical features :

- Wheezing.
- Stridor.
- Complete block : Collapse.

Dx : X-ray.

mx : Bronchoscopic removal.



## Thoracoscopes :

- 9 criteria
- Prognostic score of mortality post lung resection.

FEV<sub>1</sub>

DLCO (Diffusion capacity)

} Prediction of dyspnoea post lung sx.

## Spontaneous Pneumothorax & Empyema

00:04:44

### SPONTANEOUS PNEUMOTHORAX

Air in pneumothorax.

#### Types :

Primary spontaneous pneumothorax :

- Young people, tall.
- males > females.
- usually family history present.
- Leak from blebs : usually upper lobe.
- Lung function : Normal

↓  
Better pneumothorax tolerance.

Secondary spontaneous pneumothorax :

- Secondary to underlying lung disease.
- TB, tumors, emphysema.
- Older patients.
- Pneumothorax tolerance is less as compared to primary.

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**Clinical features :**

- Sharp pleuritic pain.
- Partial collapse.
- Complete collapse.

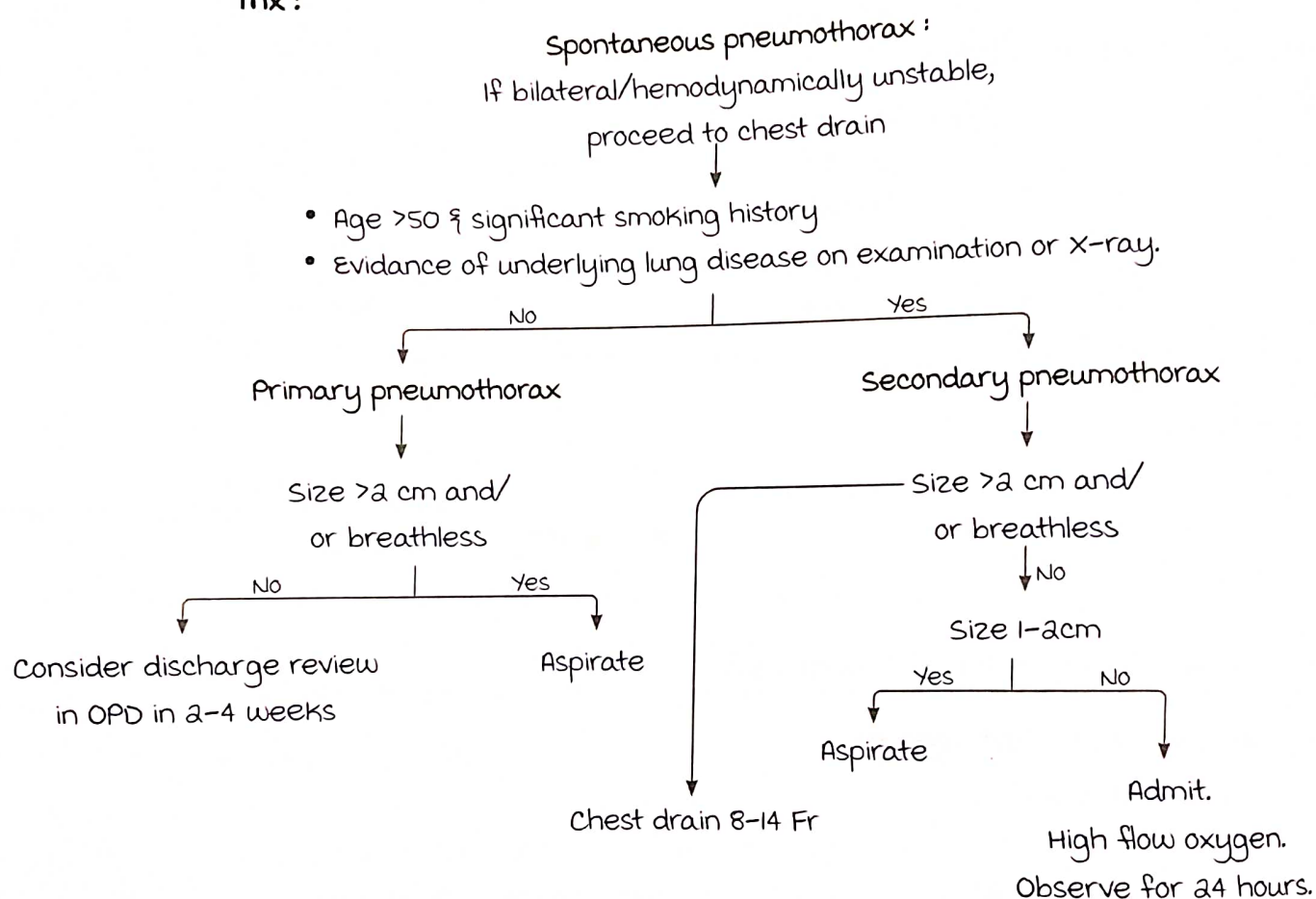
O/E :

- Hyper-resonant note on percussion.
- Absent breath sounds.

Dx :

- Chest x-ray.

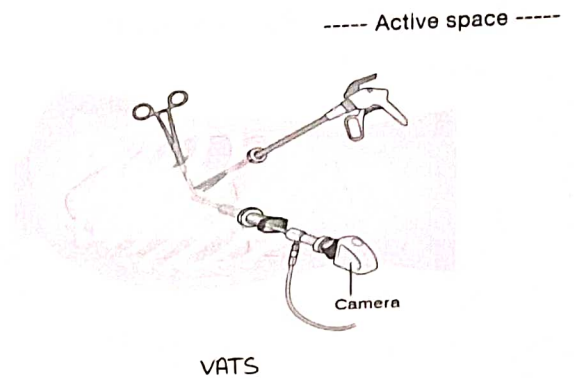
mx :

**Indications for surgical intervention in pneumothorax :**

- Secondary ipsilateral pneumothorax.
- First contralateral pneumothorax.
- Bilateral spontaneous pneumothorax.
- Pneumothorax fails to settle despite chest drainage.
- Spontaneous hemothorax : At-risk professions such as pilots and divers.
- Pregnancy.

**Interventions :**

- Pleurodesis : TALC/Tetracycline.
- Strip pleura using VATS (Video Assisted Thoracoscopic Sx)  
Space created by : **Collapsing one lung**  
↓ via  
Double lumen endotracheal tube.
- ICD.

**EMPHYEMA**

Accumulation of pus in pleural space.

**Causes :**

|                                |                                                                                                                                                                                      |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pulmonary infection            | <ul style="list-style-type: none"> <li>• Unresolved pneumonia.</li> <li>• Bronchiectasis.</li> <li>• Tuberculosis.</li> <li>• Fungal infections.</li> <li>• Lung abscess.</li> </ul> |
| Aspiration of pleural effusion | <ul style="list-style-type: none"> <li>• Any etiology.</li> </ul>                                                                                                                    |
| Trauma                         | <ul style="list-style-type: none"> <li>• Penetrating injury.</li> <li>• Surgery.</li> <li>• Esophageal perforation.</li> </ul>                                                       |
| Extrapulmonary sources         | <ul style="list-style-type: none"> <li>• Subphrenic abscess.</li> </ul>                                                                                                              |
| Bone infection                 | <ul style="list-style-type: none"> <li>• Osteomyelitis of ribs or vertebrae.</li> </ul>                                                                                              |

**Phases :****1. Exudative phase :**

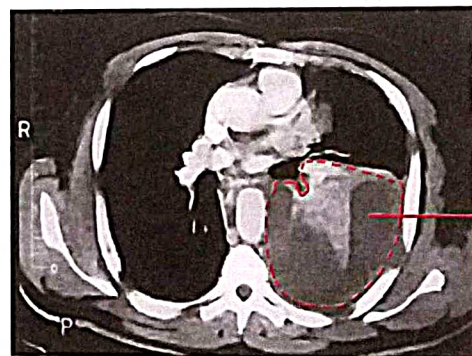
- O/E : Fever & pus.
- Imaging :  
CXR/CT : Collection of pus on one side.
- Cause :  
**Staph. aureus** > Gram -ve bacteria.
- mx : Aspiration + antibiotics.

**2. Fibrinopurulent phase :**

- O/E : Thickening of pus & pleura.
- mx : **Chest tube insertion** (Aspiration is difficult).

**3. Organized phase :**

- O/E : Thickened pleural entrapment of lung.
- mx : **Stripping of pleura/VATS + decortication.**



Empyema :  
Fibrinopurulent  
stage

----- Active space -----

## Lung Cancer

### RISK FACTORS

- Smoking.
- Pollution.
- Asbestos exposure.

Note :

- Only 15-20% patients → Amenable to surgical resection.
- Immunotherapy : Revolutionised Rx of lung cancer.

### TYPES

#### 1. Small cell (Oat cell) lung cancer (SCLC) :

- Strongest association with smoking.
- males > females.
- Poor prognosis.
- Highly sensitive to chemotherapy
- maximum number of paraneoplastic syndromes.

| Paraneoplastic syndrome | Hormone/antibody               |
|-------------------------|--------------------------------|
| Cushing's disease       | Adrenocorticotrophic hormone   |
| SIADH                   | Antidiuretic hormone           |
| Acromegaly              | Growth hormone-related peptide |
| Lambert-Eaton syndrome  | Anti-VGCC                      |

SIADH : Syndrome of inappropriate secretion of antidiuretic hormone.

VGCC : voltage gated calcium channel.

Pathology :



Deposition of DNA around blood vessels  
 ↓ On staining  
 Blue/purple staining around blood vessels

Aszopardi effect &  
 Salt & pepper appearance

IHC markers :

- Chromogranin.
- NCAM (Neural cell adhesion molecule).

## SVC syndrome :

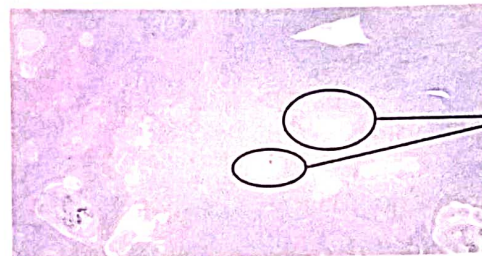
- Oncological emergency.
- Clinical features :
  - Facial & cerebral edema.
  - $\pm$  seizures.
- Dx : CECT.
- mx :
  - Radiotherapy (Shrink tumor).
  - Steroids ( $\downarrow$  intracranial pressure).

----- Active space -----

## a. Non small cell lung cancer (NSCLC) :

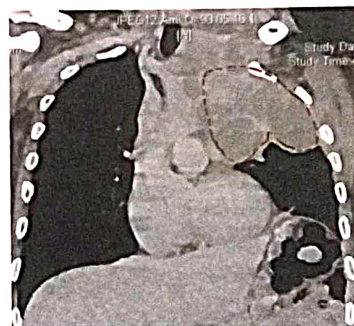
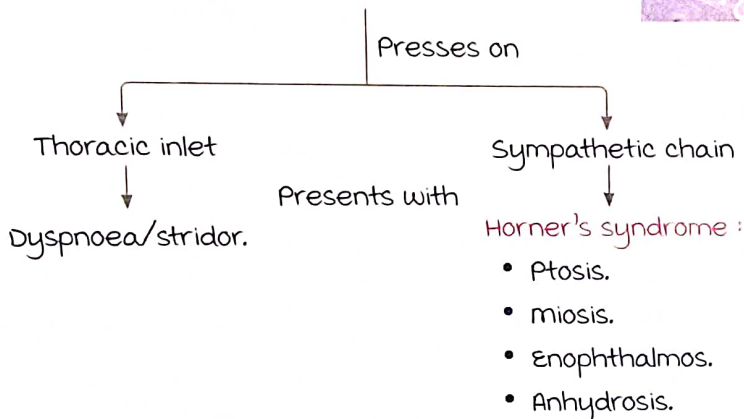
## a. Squamous cell carcinoma (SCC) lung :

- Centrally placed.
- Associated with smoking : m/c lung cancer in smokers.
- Can lead to pancoast tumor.



Keratin pearls

SCC lung



Pancoast tumor

- mx : Radiotherapy + immunotherapy.

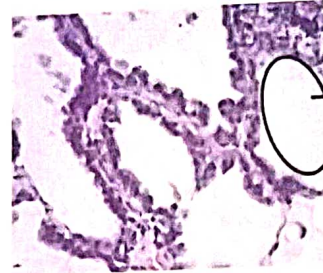
## b. Adenocarcinoma :

- Females > males.
- m/c lung cancer overall.
- Peripherally placed + slow growing.
- Develops in pre-existing lung cavities. (D/t TB/brochiectasis).
- Early metastasis.
- IHC marker : NAPSIN 4.
- Associated with mutation : ALK, RAS.

----- Active space -----

## c. Bronchoalveolar carcinoma/Adenocarcinoma in-situ

- Insitu cancer : Good prognosis.
- multifocal.
- usually resectable.
- Spread : Along broncho-alveolar lining → **Lepidic pattern**.



Filigree/Lepidic pattern

Adenocarcinoma in-situ

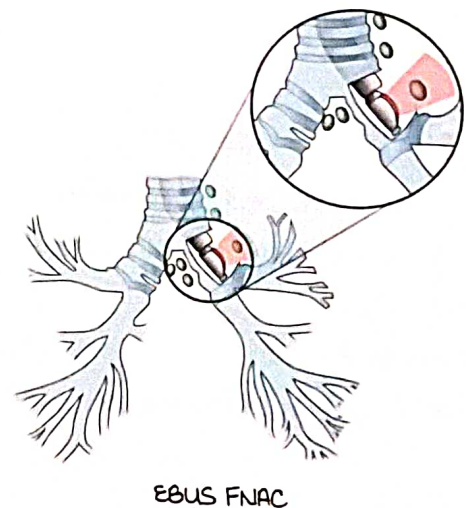
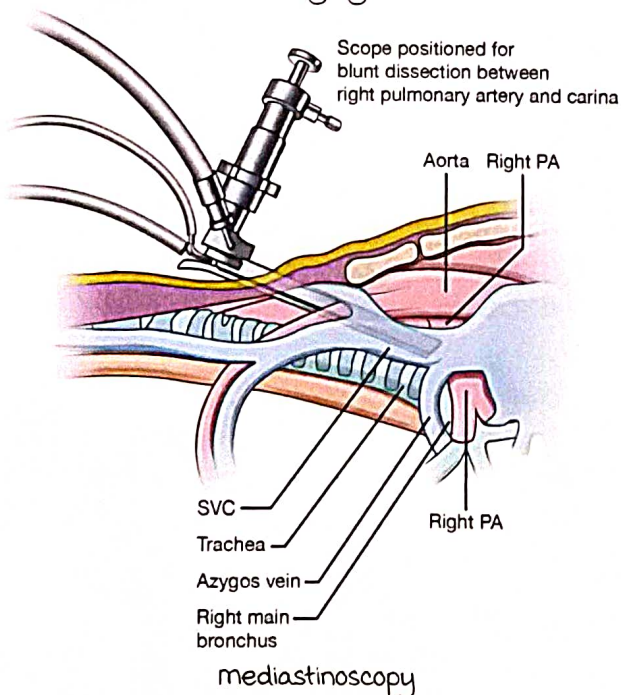
## CLINICAL FEATURES

- Symptoms :
  - m/c : **cough**.
  - Hemoptysis.
  - Pain.
  - Weight loss.
- metastasis :
  - m/c **symptomatic** : Brain.
  - very common : Adrenal gland (Asymptomatic).
  - Bones.
  - Lymphatic spread.

## WORK-UP

## Investigations :

- CXR : Initial.
- HRCT (High resolution CT) : Extent of tumor.
  - IOC.
- Biopsy : Type of tumor.
  - CT guided biopsy.
  - mediastinoscopy →
    - Check operability.
    - Take biopsy.
  - EBUS FNAC (Endobronchial US guided FNAC).
- PET-CT : Staging IOC.



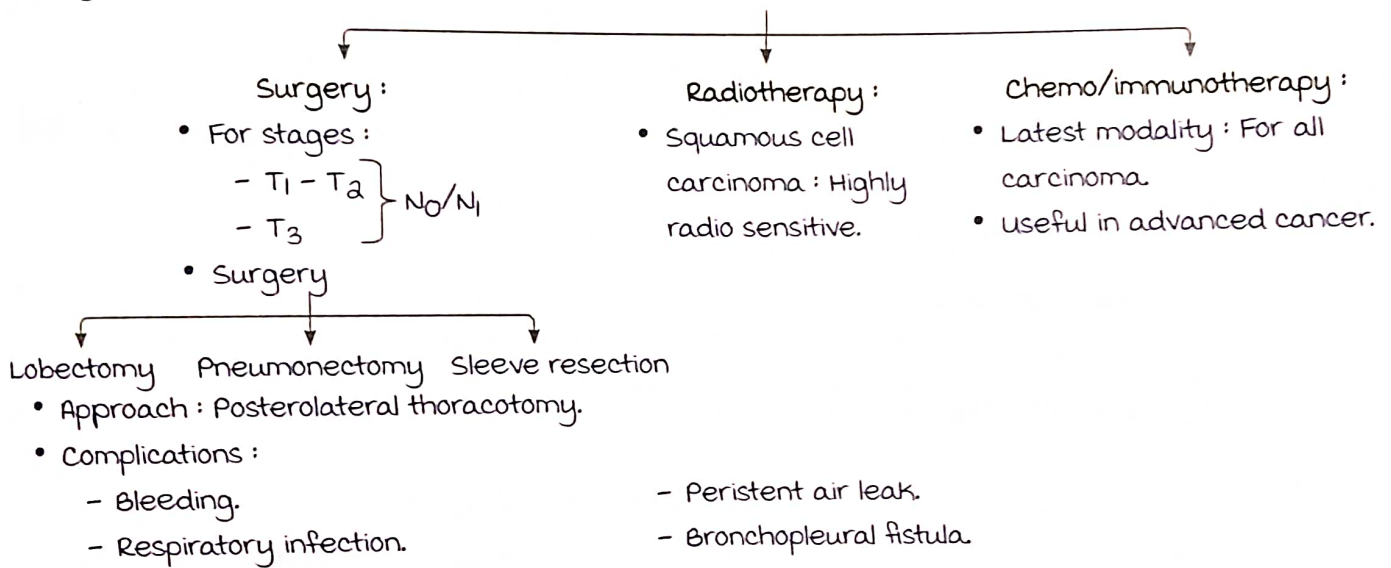
EBUS FNAC

## staging:

----- Active space -----

|                     |                                                                                                                                                                  |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T <sub>1</sub>      | ≤3 cm surrounded by lung/visceral pleura, not involving main bronchus.                                                                                           |
| T <sub>1a(mi)</sub> | minimally invasive carcinoma.                                                                                                                                    |
| T <sub>1a</sub>     | ≤1 cm.                                                                                                                                                           |
| T <sub>1b</sub>     | >1 to ≤2 cm.                                                                                                                                                     |
| T <sub>1c</sub>     | >2 to ≤3 cm.                                                                                                                                                     |
| T <sub>2</sub>      | >3 to ≤5 cm or<br>Involvement of main bronchus without carina or invasion of visceral pleural or atelectasis or post obstructive pneumonitis extending to hilum. |
| T <sub>2a</sub>     | >3 to ≤4 cm.                                                                                                                                                     |
| T <sub>2b</sub>     | >4 to ≤5 cm.                                                                                                                                                     |

## management:



## SUMMARY

| Feature                  | SCC                     | Adeno Ca                                                         | Small cell Ca                                               | Large cell Ca |
|--------------------------|-------------------------|------------------------------------------------------------------|-------------------------------------------------------------|---------------|
| Incidence                | m > F                   | F > m                                                            | m > F                                                       | m > F         |
| Location                 | Central                 | Peripheral                                                       | Central                                                     | Peripheral    |
| Smoking association      | Smoking association     | Non smokers                                                      | Strongest association with smoking                          | Both          |
| Paraneoplastic syndromes | Hypercalcemia           | migratory thrombophlebitis                                       | Cushing's syndrome, SIADH                                   | Gynecomastia  |
| Pathogenesis             | PS3                     | K ras, EGFR, ALK                                                 | L-myc                                                       |               |
| Precursor lesion         | Carcinoma in situ (CIS) | Adenocarcinoma in situ<br>Atypical adenomatous hyperplasia (AAH) | DIPNEH (Diffuse idiopathic pulm neuroendocrine hyperplasia) | -             |

---- Active space ----

| Feature | SCC                        | Adeno Ca                          | Small cell Ca                                                            | Large cell Ca            |
|---------|----------------------------|-----------------------------------|--------------------------------------------------------------------------|--------------------------|
| H & E   | Keratin pearls, desmosomes | Glands lined by pleomorphic cells | Small cells, salt & pepper chromatin, nuclear moulding, Azzopardi effect | Large, pleomorphic cells |
| IHC     | CK, P63, P40               | TTF 1, NAPSIN A                   | NSE, chromo, synapto                                                     | -                        |

## Benign Lung Tumors

00:34:37

### HAMARTOMA

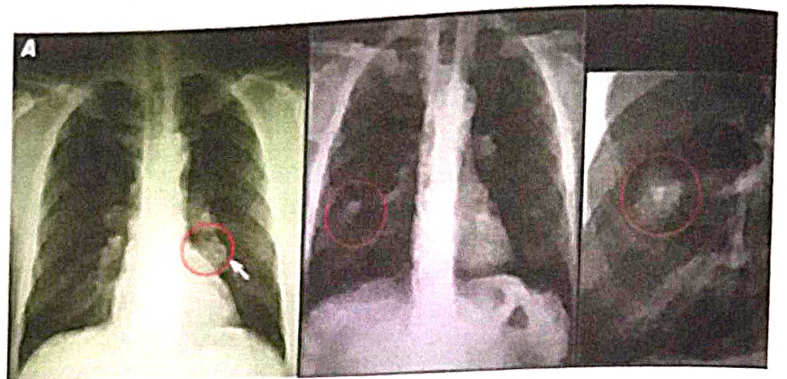
m/c benign lung tumor.

#### Clinical features :

- Asymptomatic.
- Cough.
- Hemoptysis.

#### Investigations :

- CXR : Coin shaped lesions.
- CT : Confirmatory.

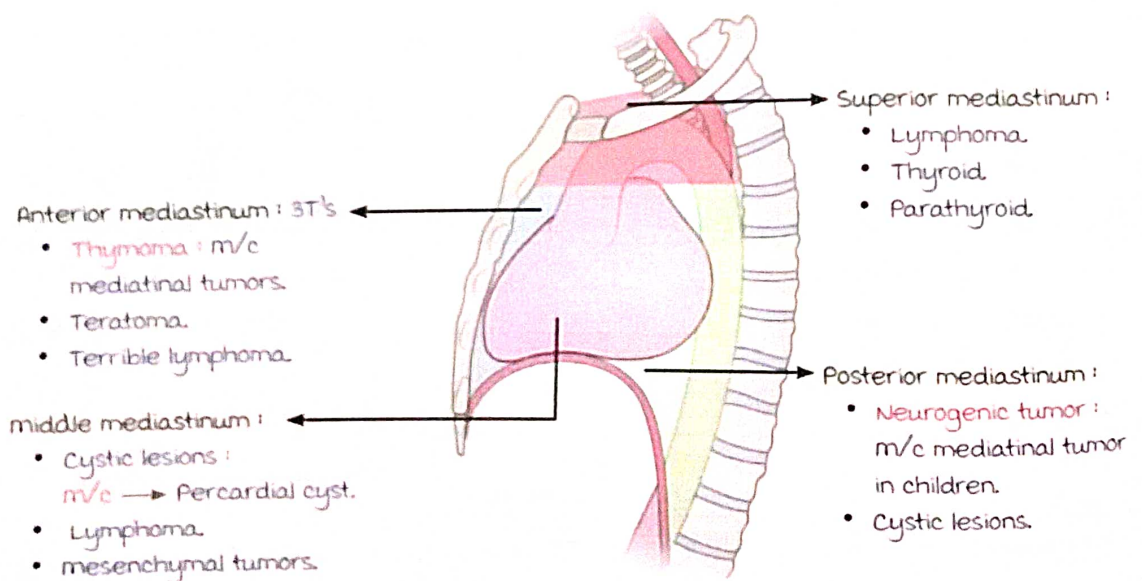


Coin shaped lesion

#### management :

VATS guided excision.

## MEDIASTINAL TUMORS



**Thymoma :**

- m/c mediastinal tumor overall.
- Site : Anterior mediastinum.
- usually benign.
- A/w myasthenia gravis.

**management :**

- Excision :
  - Small lesion : Transcervical excision.
  - Large lesion : VATS/Thoracotomy.

**masoka thymoma staging :**

| Stage | Description                                                                                                                                                                     |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I     | <ul style="list-style-type: none"> <li>• macroscopically completely encapsulated.</li> <li>• microscopically no capsular invasion.</li> </ul>                                   |
| II    | <ul style="list-style-type: none"> <li>• macroscopic invasion into surrounding fatty tissue or mediastinal pleura.</li> <li>• microscopic invasion into the capsule.</li> </ul> |
| III   | macroscopic invasion into neighboring organs<br>(Pericardium, great vessels, lungs).                                                                                            |
| IVA   | Pleural or pericardial dissemination.                                                                                                                                           |
| IVB   | Lymphogenous or hematogenous metastasis.                                                                                                                                        |

**Germ cell tumors :**

- m/c extra gonadal site : Anterior mediastinum.
- males > females.
- If benign : Teratoma.
- If malignant : ↑ tumor markers ( $\alpha$ FP, LDH,  $\beta$ -hCG).
- mx : Excision

↓ If malignant  
Chemotherapy

----- Active space -----

## COMMON SURGICAL SWELLINGS

## Sebacaceous Cyst

00:01:02

Blocked hair follicle duct.

Clinical features :

Sites : Anywhere **except palms and soles** (No hair follicles).

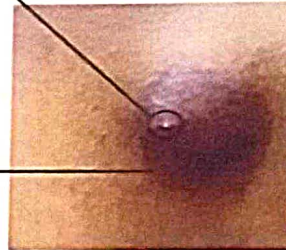
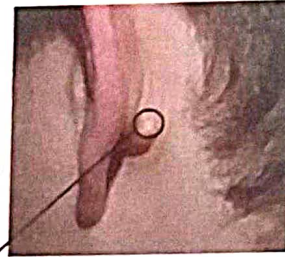
- Swelling where skin cannot be pinched (Arises from skin).
- whitish discharge.
- Pain (Sometimes).
- whitish centre : **Punctum** (Characteristic).

Complications :

1. Inflamed sebaceous cyst (Infected).
2. Multiple sebaceous cysts : Seen in scrotum and scalp.
3. **Sebaceous horn** : Secretions harden to form a horn like structure.

management : Excision of cyst.

Punctum



Scrotal sebaceous cysts

## Dermoid Cyst

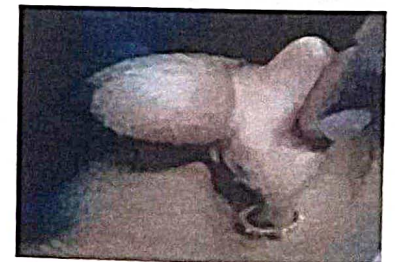
00:03:50

Pathophysiology : Formed when **epithelial tissue** is trapped along the **lines of embryonic fusion**.

Sites : Outer canthus of eye, Post auricular region.

Clinical features :

- Cystic swelling (Fluctuation +ve).
- **Skin can be pinched** over the swelling.



Investigations : X-ray/CT must be done to rule out intracranial extension before Sx.

management : Surgical excision.

Implantation dermoid :

- Dermoid which occurs due to injury.
- **m/c** after a ear piercing.
- mx : Surgical excision.



## Lipoma

00:06:26

----- Active space -----

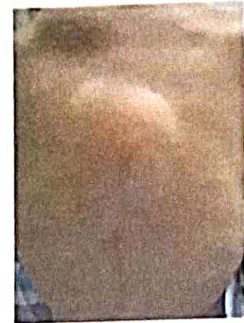
encapsulated collection of fat (m/c swelling in the body).

Clinical features :

- Skin can be pinched over swelling (Arises from subcutaneous tissue).
- **Pseudofluctuation** (Swelling expands only in one axis).
- **Slip sign (+)**.
- **Dercum's disease** : multiple lipomas/lipomatosis (Benign condition → Observation).

management :

- If symptomatic (Painful) or large : Excision.
  - Lipoma in :
    - Retroperitoneum.
    - Thigh.
    - B/w shoulder blades.
- High potential for sarcomatous change (must be observed closely).



Excised lipoma.

## Cervical Lymphadenopathy

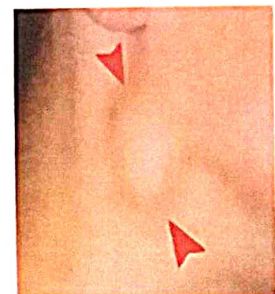
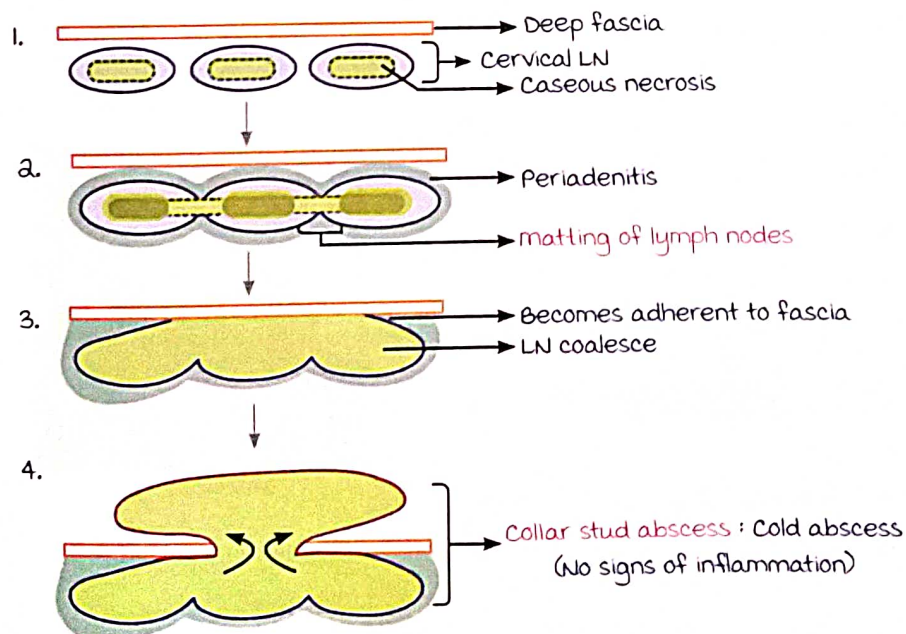
00:09:08

Cervical lymph node enlargement due to pathological or reactive causes.

Etiology : m/c pathological cause in India is **tuberculosis**.

**Tubercular cervical lymphadenopathy :**

Pathophysiology : Tubercular bacilli infect the cervical LN.

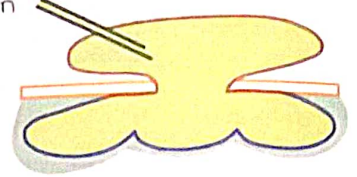


----- Active space -----

**Diagnosis :**

- Cold abscess in cervical region.
- Fluctuant swelling.
- Confirmation : **Antigravity aspiration** → Ziehl - Neelsen staining  
(If aspirated from below, sinus tract can form).

Antigravity aspiration



management : Anti-tubercular therapy.

**Carbuncle**

00:13:26

multiple small abscesses coalesce **To form** → Large abscess.**Clinical presentation :**

- **m/c** in diabetics.
- Has multiple pus points.
- Site : **Nape of the neck** (m/c).



management : Drainage with a cruciate incision (To drain all the abscesses).

**Branchial Cyst and Branchial Sinus/Fistula**

00:14:32

**Branchial Cyst :****Pathophysiology :**

- Fusion of 2<sup>nd</sup> arch with 6<sup>th</sup> arch forms cervical sinus.
- Normally cervical sinus obliterates.
- If it persists → Forms branchial cyst.

**Site :**

- Swelling over neck.
- Anterior border of the sternocleidomastoid (SCM).
- Junction b/w **upper and middle 1/3<sup>rd</sup>**.

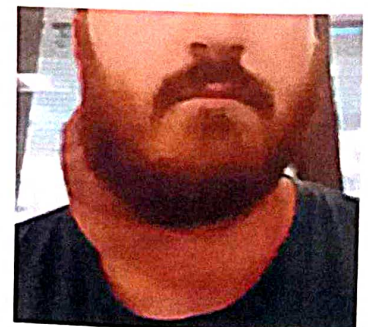
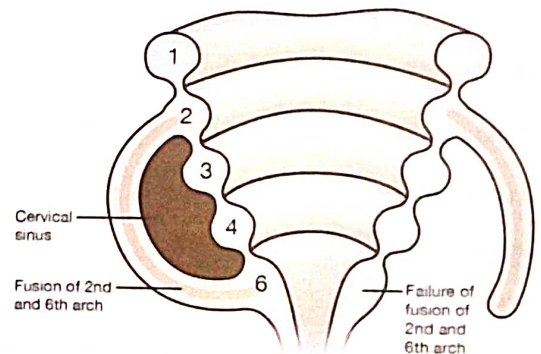
**Clinical features :**

- Fluctuation (+) : Cystic swelling.
- Transillumination +ve.

management : FNAC → Surgical Excision.

**Branchial sinus/Fistula :**

- Formed by failure of fusion of 2<sup>nd</sup> and 6<sup>th</sup> arch.
- Site : Anterior border of SCM → b/w **middle and lower 1/3<sup>rd</sup>**.
- mx : Excision.



Branchial Cyst

# COMMON ULCERS

----- Active space -----

## Ulcers

00:01:00

Breach in the continuity of epithelium/mucosa.

## TYPES

|                     | Venous ulcer | Arterial ulcer      | Trophic ulcer          | Diabetic ulcer         |
|---------------------|--------------|---------------------|------------------------|------------------------|
| Site                | Gaiter area  | Dorsum/lateral side | Sole/base of great toe | Sole/base of great toe |
| Arterial pulsations | Normal       | Absent pulsations   | Normal pulsations      | may be absent          |
| Dilated veins       | Present      | -                   | -                      | -                      |
| Sensations          | Normal       | Painful             | ↓ sensations           | ↓ sensations           |
| margin              | Sloping      | Punched out         | Punched out            | Punched out            |

### varicose ulcers :

Site : medial malleolus (m/c).

Features :

- Shallow ulcer.
- Pale granulation tissue.
- Pigmented margin (D/t hemosiderin).
- Non-healing ulcer.



venous ulcer



varicose ulcer

### marjolin's ulcer :

Squamous cell cancer.

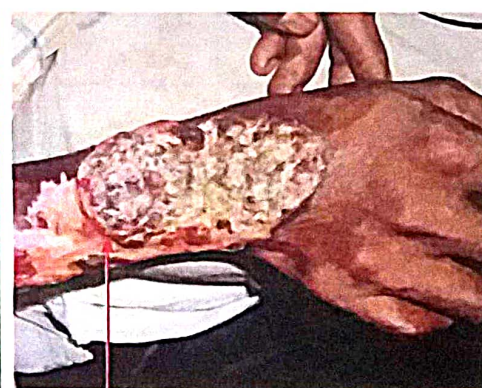
Risk factors :

- Long standing venous ulcer.
- Burn scars.

Features : Raised, everted margins.

management : Wide local excision.

C/I : Radiotherapy (D/t scarring & recurrence).



Burn scar  
marjolin's ulcer

----- Active space -----

**Arterial ulcers :****Features :**

- Loss of muscle mass/hair.
- Shiny skin.



Arterial ulcer

**Trophic ulcer :****Risk factors :**

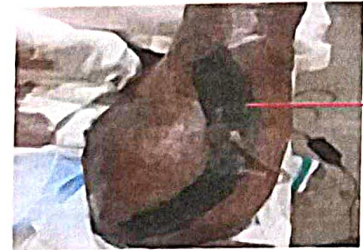
- Paralyzed patients.
- Diabetics.



Trophic ulcer

**Diabetic ulcer****Predisposing factors :**

- microangiopathy (Arterial disease).
- ↑ glucose.

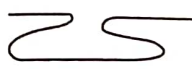


VAC dressing

**Stages :**

| Stages     | Stage 1                                                                                                               | Stage 2                                                                                                                                              | Stage 3                                                                                            | Stage 4                                                                                                        |
|------------|-----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
|            | Inflammation but no breach<br>     | Superficial ulcer<br>                                             | Deep ulcer<br> | Ulcer + osteomyelitis<br> |
| management | <ul style="list-style-type: none"> <li>• Antibiotics.</li> <li>• Diabetic control.</li> <li>• Off loading.</li> </ul> | <ul style="list-style-type: none"> <li>• Antibiotics</li> <li>• Diabetic control</li> <li>• Off loading + debridement &amp; VAC dressings</li> </ul> |                                                                                                    | Amputation                                                                                                     |

## ULCER EDGES

1.  1. Sloping :
  - Healing.
  - Venous.
2.  2. Punched out :
  - Arterial.
  - Neuropathic.
  - Bed sores.
  - Syphilis.
3.  3. Undermined : TB
4.  4. Rolled out, pearly white :  
BCC (Rodent ulcer)
5.  5. Raised, everted, cauliflower :  
SCC (Squamous cell carcinoma)

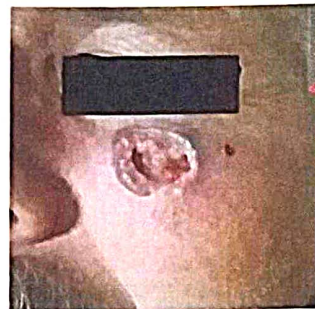
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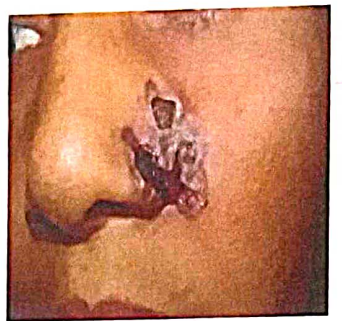
Tubercular ulcer



SCC



Basal cell carcinoma (BCC)



# SURGICAL INSTRUMENTS

## Open Surgical Instruments

00:01:00

Parts of an instrument :

- a Rings : Gripping.
- Ratchets (multiple) : Prevent wear & Tear.
- Joint :
  - Box joint.
  - Screw joint : ↑ incidences of thread getting stuck.
- Blades :
  - Fenestrated.
  - Serrated.

### 1. RAMPLEY'S SPONGE HOLDER

Fenestrated blade with serrations.

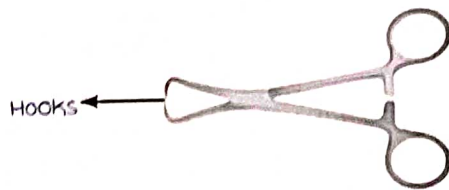
Uses : Grip sponge for part preparation.



### 2. TOWEL CLIPS

Uses :

- Hold the towels post draping.
- Pass and hold suction catheter/cautery.
- Hold tongue during tongue resection.



Mayo's towel clip

Pinch cork effect



Doyen's towel clip

### 3. BP (BRAD PARKER) HANDLE

Uses : To mount blades.

Holding Position :

- Like a paint brush.
- Palming grip.



#### 4. YANKAUER'S SUCTION CANNULA

uses : Suction blood/fumes during Sx.

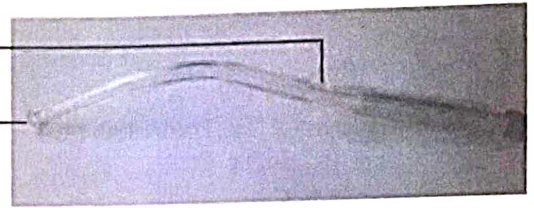
Types :

- Plastic.
- metallic.

Suction creating hole

Suction end

----- Active space -----



#### 5. FORCEPS

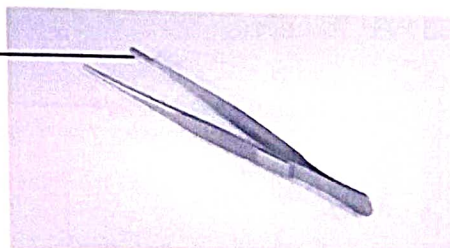
Spring loaded.

uses : To hold tissues.

Types :

- Plain.
- Toothed.
- vascular.
- Special.

Transverse serrations without teeth  
↓  
Relatively atraumatic



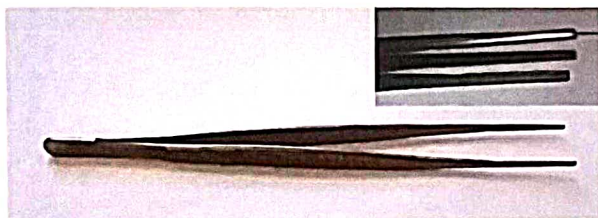
Plain forceps

Teeth (+)

- Gripping tough structures.
- Traumatic.

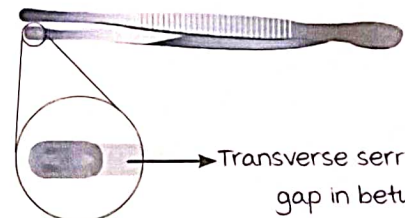


Toothed forceps



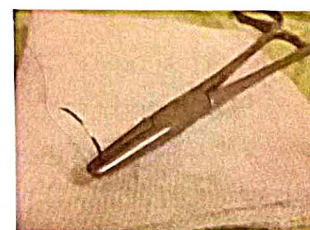
DeBakey's vascular forceps

minute serration  
+  
Raised edges  
↓  
Grip vascular structures.



Russian Forceps

Criss Cross serrations :  
Prevent needle movement.



Needle holder

#### 6. NEEDLE HOLDER

uses : Grip needle.

usage :

- Hold needle at 1/3<sup>rd</sup>-2/3<sup>rd</sup> junction from swaged and pointed end respectively.
- Enter at 90° to skin.
- Toothed forceps to grip needle while suturing.

#### 7. SCISSORS

a. Heavy (Mayo scissor) :

- Screw joint + Blades.
- uses : Cutting sutures/sheaths.



----- Active space -----

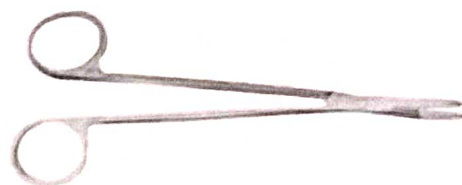
## b. metzenbaum Scissors :

- Lighter than Mayo's.
- AKA Tissue dissecting scissor.
- Avoid suture cutting.



## c. McIndoe Scissors :

- Finer than metzenbaum.
- Uses :
  - Cut fine tissues.
  - Ophthalm/ENT surgeries.



Note : Instruments used in skin suturing

- Needle holder.
- Toothed forceps.
- Scissors.

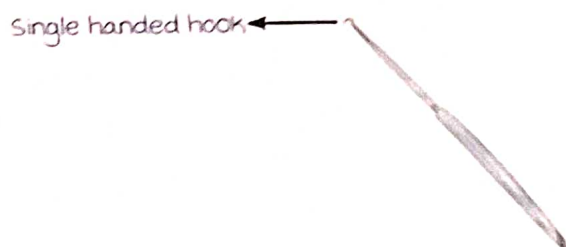
## 8. RETRACTORS

## a. Non-self retaining :

Need to be held.

## (i) Skin hooks :

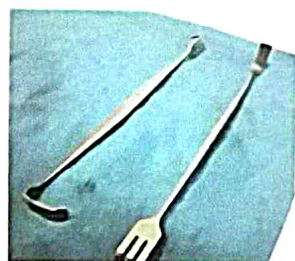
- Hold raised flaps in mastectomy, thyroidectomy, parotidectomy.
- Single/double handle hooks.



Skin hooks

## (ii) Cats paw retractor :

Retract superficial tissue.



Cats paw Retractor

## (iii) Langenbeck retractor :

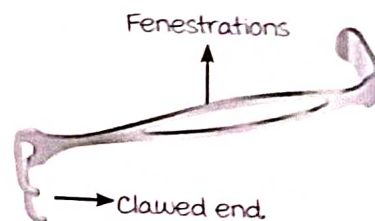
- Right angle retractor.
- Retract superficial tissue.
- Fenestration (+) : Instrument lighter.
- Broader retractor : Cholecystectomy.



Langenbeck retractor

## (iv) Army navy/Czerney's retractor :

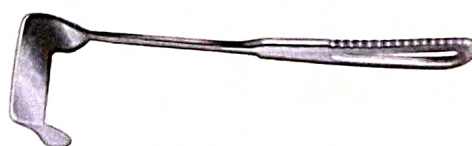
- One end : 90°.
- Other end : Clawed with gap.
- Use : Suturing tissues while retracted.



Czerney's retractor

## (v) Morris retractor :

- Broader than Langenbeck's.
- Advantages over Langenbeck's :
  - Prevents slipping away of structures.
  - Better grip & lighter.



Morris retractor

## (vi) Doyen's retractor :

- use : Bladder retraction.
- Advantages : ↑ Convexity → Easy retraction.

## (vii) malleable retractor :

use : Commonly used in abdominal surgeries for bowel retraction.

## b. Self-retaining retractors :

## (i) mastoid retractors :

- Superficial tissue retractor.
- Retract tissue Lock instruments → Self retraction.
- use : **Thyroid/parotid surgeries.**

## (ii) Balfour self retaining retractor :

- Deeper abdominal structures.
- Parts :
  - a. Bar : Inserted in abdomen & adjusted.
  - b. Blades : For retraction.
  - c. Space to attach Doyen's retractor.

## (iii) Bookwalter retractor :

- Complex abdominal malignancy Sx.
- Parts :
  - i. Ring : Attached to operating table.
  - ii. Space for attachment of multiple retractors.

## 9. ARTERY FORCEPS

## a. Spencer wells forceps :

- Straight artery forceps.
- use : Hold bleeders during Sx.
- Comes in various sizes.

## b. Curved artery forceps :

- use : Hold bleeders.
- Advantage : Better visualization.
- Smallest artery forcep : mosquito forcep.

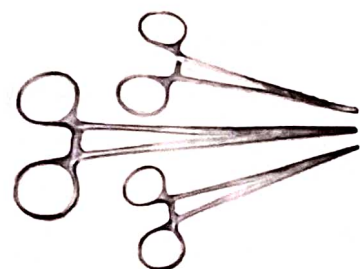
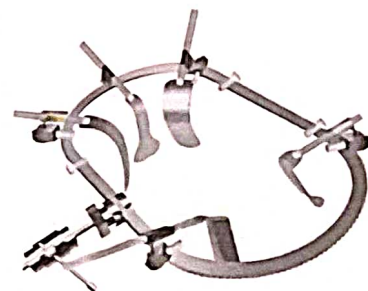
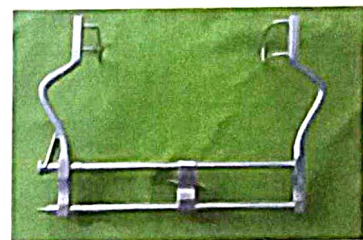
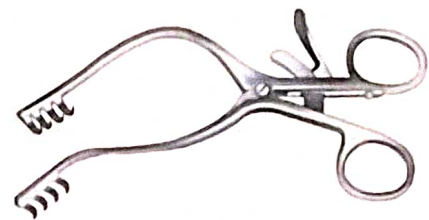
----- Active space -----



Doyen's retractor



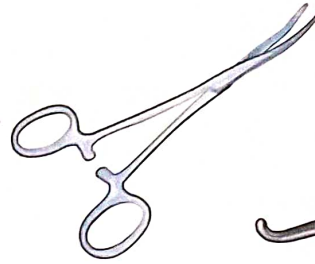
malleable retractor



----- Active space -----

**10. SPECIAL FORCEPS****a. Kelly's forceps :**

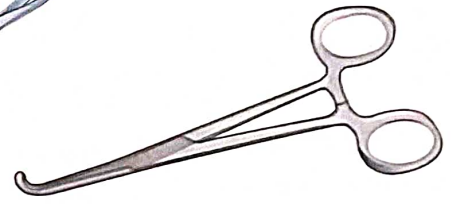
- Curved, much bigger forceps.
- more gradual curve.
- Use : Abdominal surgeries.



kelly's forceps

**b. Right angle forceps :**

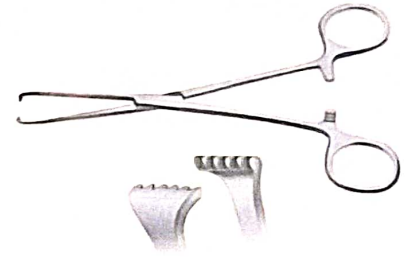
- Right angle curve.
- Use : Tie pedicles.



Right angled forceps

**c. Allis forceps :**

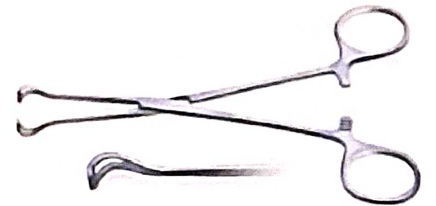
- Blades with serration & teeth → Traumatic.
- Use : Hold tough structures → Sheath/fascia.



Allis forceps

**d. Babcock forceps :**

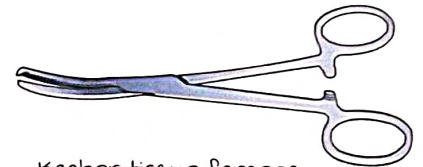
- Gap b/w two fenestrated ends.
- Use : Hold tubular structures (Appendix/vas/fallopian tubes).
- Advantages : Does not crush structures.



Babcock's forceps

**e. Kocher tissue forceps :**

- Interlocks → Crush tissue
- Use :
  - Hold tough structures.
  - Hold cervix during hysterectomy.



Kocher tissue forceps

**Laparoscopic Instruments**

00:26:45

**Laparoscopic instruments :**

- Long : Operated at depth.
- Insulated : Prevent current leaking out from cautery.

**1. Veress needle :**

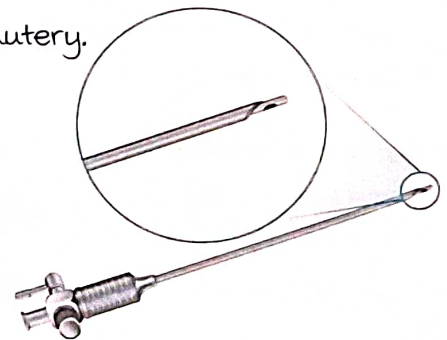
- Closed method of creating pneumoperitoneum.
- Spring loaded needle.
- Parts :

- Outlet/inlet valve.

- Blunt tip in front

On pressing  
Bevelled tip

Prevents tissue injury



## 2. Laparoscopic port/trocar :

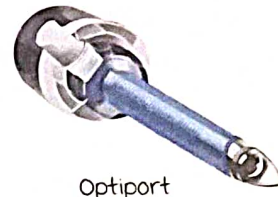
1<sup>st</sup> port : Inserted blindly.

Types :

- Sharp.
- Blunt : **Hassan's trocar**.
- Bladeless : **Optiport**  
(Better visualization).

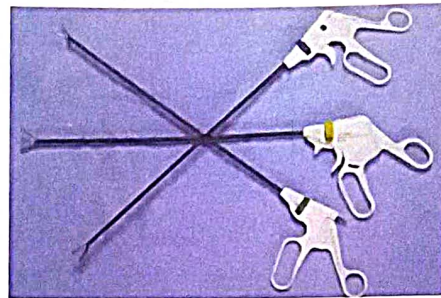


Dilating tip bladeless trocar



Optiport

----- Active space -----



Maryland's dissector

## 3. Maryland's dissector :

- Curved, serrations like artery.
- Workhorse of lap Sc.

## Thoracotomy Instruments

00:31:00

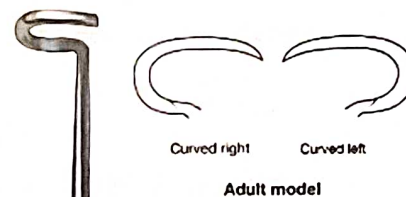
### 1. Periosteal elevator :

- AKA : Farabeuf's periosteal elevator.
- Use : Raise periosteum over rib.

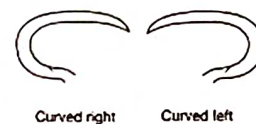


### 2. Rib raspatory :

- One for each side.
- Used to resect periosteum & muscles.
- Position :
  - Handle : Lateral.
  - Convexity : Upwards.
  - Tip : Downwards.



Adult model

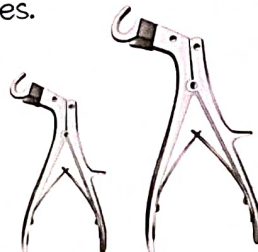


Children model

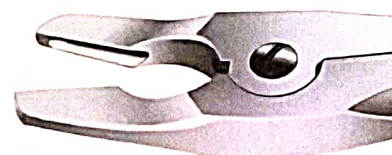
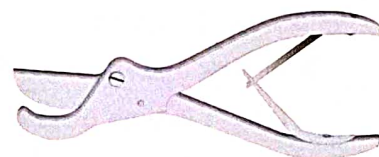
### 3. Rib cutter :

Blunt lower edge : Prevent pleural injury.

- Bone cutter : Sharp edges.



Doyen's rib raspatory



### 4. Bone nibbler :

Nibble off sharp edges after cutting bone.

----- Active space -----

**5. Lung retractor :**

Fenestrated blade : Allows lung expansion on retraction.



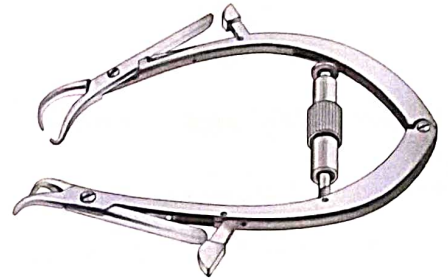
Allison lung retractor

**6. Lung forceps :**

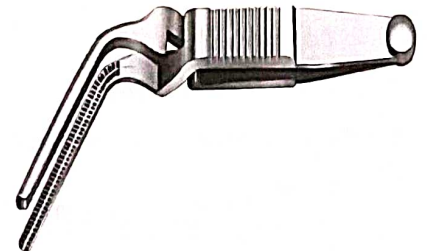
- Resection of small tissues : Hamartomas/nodules.
- Triangular end : Provides bloodless field.

**7. Joll's thyroid retractor :**

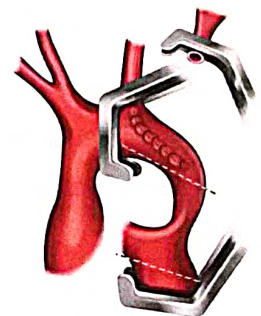
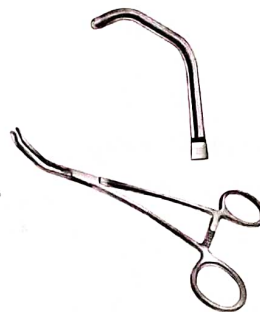
- Obsolete.
- 2 retractors on either side to retract thyroid.

**VASCULAR SURGICAL INSTRUMENTS****1. Clamps :****a. Bulldog clamp :**

- Self retaining clamp with pinch cork effect.
- Serrations (+).
- Advantage : Clamp temporarily without crushing structures.

**b. Satinsky vascular clamp :**

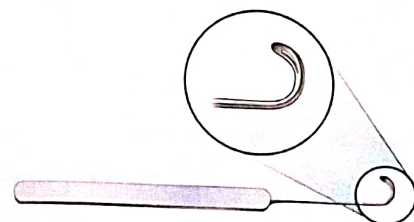
- Curved clamp with serrations.
- Use : Over pedicle.
- Advantage : Better visualization.



Clamping of aorta

**c. Aneurysm needle :**

- J hook with opening at end.
- Use : Ligation of aneurysm.

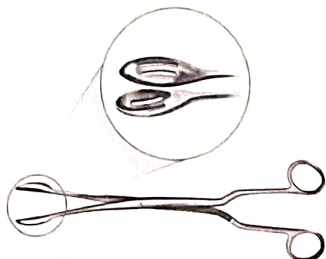


## OTHER INSTRUMENTS

----- Active space -----

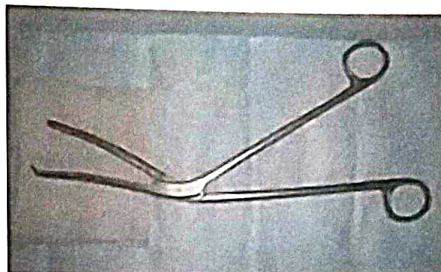
## 1. Ovum forceps :

- 2 convex ends with fenestrations.
- Use : Remove retained placental/ fetal tissue.



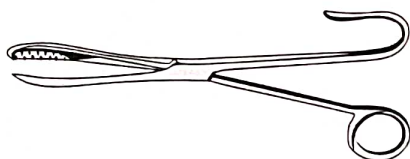
## 2. Cheatle's forceps :

- 2 ends without lock.
- Use : Pick up gauze piece.



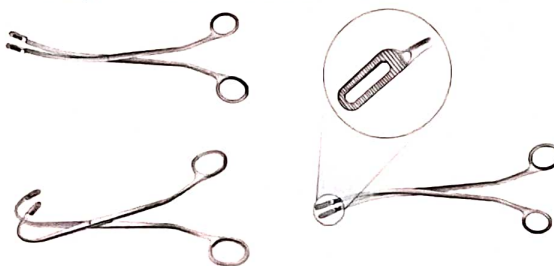
## 3. Cystolithotomy forceps :

- One complete ring for thumb.
- One open part : Other 4 fingers.
- No lock : Prevent crushing of stone.
- Studs at end : Better grip.



## 4. Pyelolithotomy forceps :

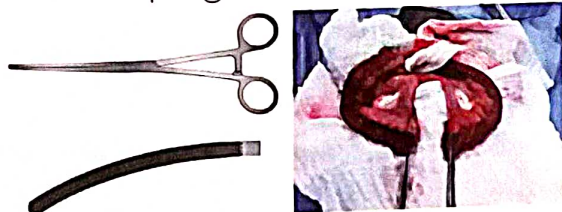
- Obsolete.
- Varied angulations : Better calyceal access.
- Ends : Transverse serrations without lock.



## 5. Bowel clamps :

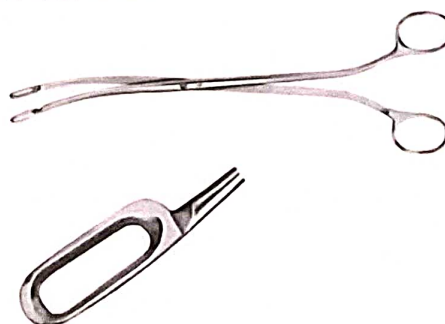
## a. Non crushing clamp :

- Doyen's Intestinal Clamp.
- Longitudinal serrations : Prevent tissue crushing.
- Prevent spillage of fecal matter.

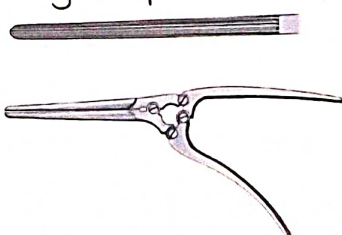


## 6. Choledocholithotomy forceps :

- Desjardin's forceps.
- Use : Bring down CBD stones.
- No serrations.



## b. Payr's crushing clamp : Removes portions of bowel.



“

I don't find myself extraordinary other than the  
passion to cut and stitch.

**Dr. Devi Shetty,**  
Cardiac Surgeon

